Project Proposal

Project Samsara: A Self-Recharging Autonomous Quadcopter

Robotics and Automation Society (RAS), University of Pittsburgh
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Overview

Project Samsara builds a quadcopter that flies a GPS mission, lands, recharges, and relaunches without human servicing. "Without human intervention" means without servicing: a remote pilot is present on every flight. The long-term goal is a drone that works over terrain with no prepared infrastructure.

The project runs across two semesters:

1. **Fall 2026: autonomous flight (M4, primary deliverable).** The drone flies a takeoff, a 30 m waypoint at 10 m altitude, a return, and a landing within 2 m of the takeoff point, for five consecutive trials. M4 is due November 20, before Thanksgiving recess.
2. **Spring 2027: onboard charging electronics (M5).** The team charges the target Li-ion pack through the onboard charge path from a bench source, and logs the results. There is no ground charging pad.
3. **Spring 2027: a 3–4 cell solar test panel.** The panel measures real watts per kilogram, which feeds the design of the onboard array.

After that, the project adds the onboard solar array and the full land–charge–relaunch loop.

The vehicle uses stock ArduPilot Copter on a Pixhawk 6C, a Holybro X500 V2 frame, and a T-Motor propulsion set that has a published thrust table. RAS has an older quadcopter, but its frame is misaligned and its motors, ESCs, GPS, and flight controller have not yet been inspected. This proposal therefore prices every part as a new purchase and does not count on the old hardware; any part that passes the M1 inspection is reused and comes off the budget. Each part below is **recommended, pending team decision**. Autonomous flight (M4) costs **$838.62** in fall, inside RAS's $1,000 commitment. The full list across both semesters is **$1,424.52** before shipping and tax, and $335.32 of that is a charge-controller kit that the M3 gate may replace with a cheaper part. RAS has said it can move items between semesters.

Possible applications, once the full loop works, are persistent monitoring, disaster damage survey, and an aerial Meshtastic mesh relay. A drone that recharges itself can hold a relay position, and it still works as a relay while it is landed and charging. After a September 25 call, the Protomesh Collective has offered a Raspberry Pi Zero 2 W and nickel strip for the battery pack, has donated a spot welder for battery-pack building to the Prototype x Oakland makerspace, and has offered to discuss donating a drone.

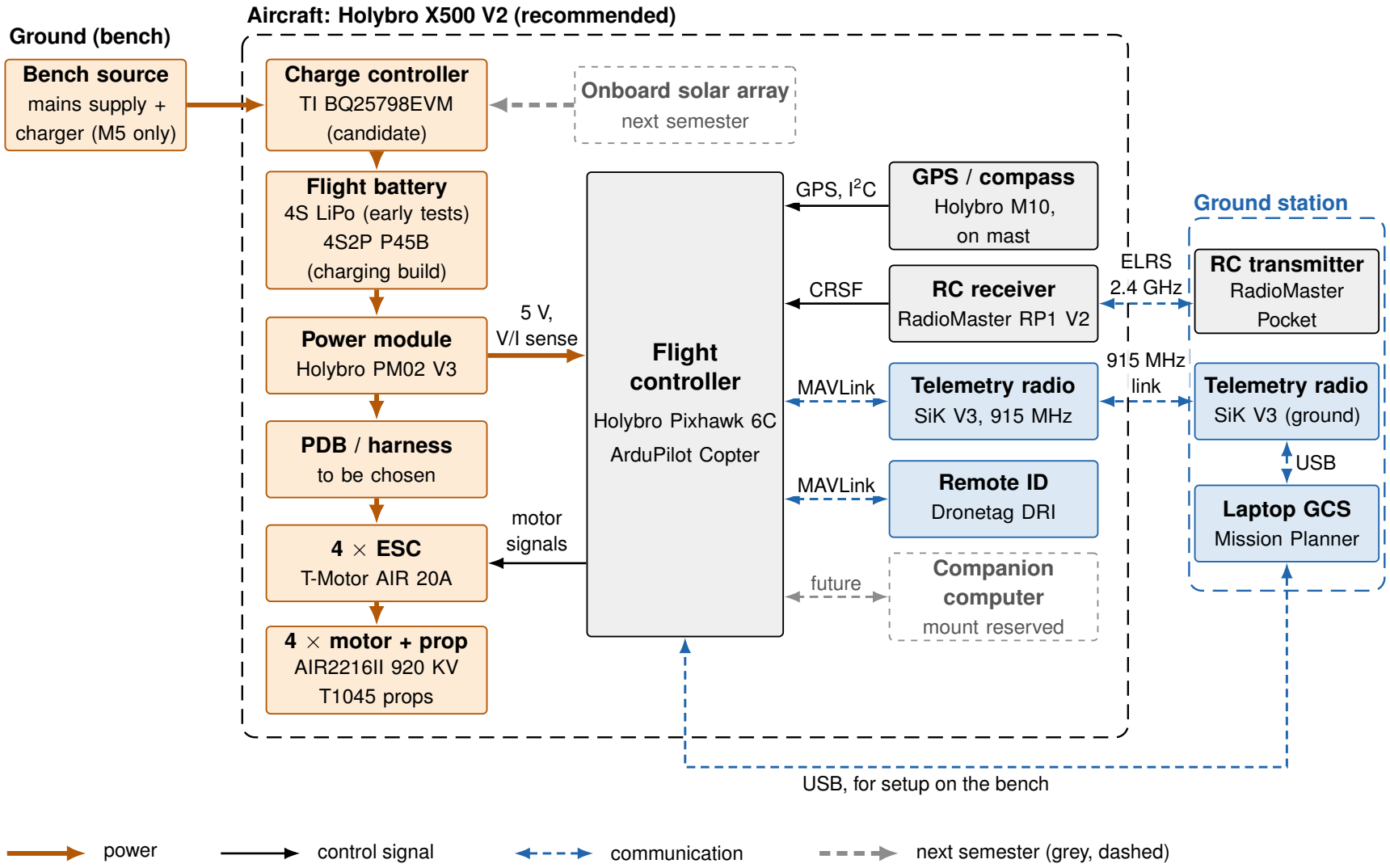


Figure 1. Thick arrows carry power, thin arrows carry control signals, and blue dashed arrows carry communication. The companion computer and the solar array are placeholders for next semester.

Design Details

System Architecture

The vehicle is a conventional ArduPilot quadcopter. Figure 1 shows the three paths that connect the blocks: power, control, and communication.

Power path. The battery connects to the power module [2]. The ESCs, or a power distribution board, connect to the module's output [2]. The Pixhawk's power port powers the Pixhawk and reads the module's voltage and current measurements [1]. With these readings, the autopilot monitors the battery and can trigger a low-battery failsafe [3]. LiPo packs are used for early flight tests. A Li-ion 21700 pack is the target for the charging build.

Control path. Each ESC connects to the autopilot output that matches its motor number [4], and each ESC drives its motor through three power leads [4]. The GPS connects to the GPS port and the compass to I^2C [5]. ArduPilot recommends a GPS mast and at least 10 cm of clearance from power wiring and batteries [5]. ArduPilot accepts PPM-Sum, SBus, and CRSF receivers, including ExpressLRS [6].

Communication path. Mission Planner is a ground control station for Copter [7]. It plans point-and-click waypoint missions and loads them into the autopilot [7]. It connects over USB or a telemetry radio [8]. A SiK radio gives a telemetry link between the autopilot and the ground station [9], in 915 MHz or 433 MHz versions [9].

Failsafes. ArduPilot can switch to RTL or Land when the RC signal is lost [19] or the battery is low [18].

Future blocks. A companion computer talks to ArduPilot over MAVLink [10]. The companion computer and the onboard solar array are next-semester work. The frame keeps a mount point and a mass allowance for the companion computer.

Component selection

Block	Recommended part	Price	Key spec	Why
Flight controller	Holybro Pixhawk 6C (plastic case)	$199.00 [37]	STM32H743, 2 MB flash [37]; 34.6 g [37]	ArduPilot Lua scripting needs at least 2 MB of flash and 80 kB of memory [31]. The 6C meets this, so next semester's relaunch logic is possible.
GPS/compass	Holybro M10 GPS	$43.99 [38]	u-blox M10, 2.0 m CEP (vendor listing) [38]; 32 g [38]	Cheapest in-stock option. The M9N lists 1.5 m CEP at $54.99 [39] and is the upgrade if M2 position hold falls short. The M8N is discontinued [36].

Block	Recommended part	Price	Key spec	Why
Frame	Holybro X500 V2 Frame Kit	$135.00 [40]	500 mm wheelbase, carbon fibre, 610 g [40]	The only frame with a published mass. The S500 V2 Frame Kit is $42.00 [41], but Holybro gives no frame-only mass for it.
Propulsion	T-Motor Air Gear 450 II combo	$129.99 [45]	4 × AIR2216II 920 KV (4S) motors, 4 × AIR 20A ESCs, 4 × T1045 props [45, 46]	T-Motor publishes a thrust table for this exact motor and prop at 16 V [45].
Power module	Holybro PM02 V3	$18.99 [42]	2S–12S, 60 A continuous, analog [42]; 20 g [42]	Analog output suits the 6C. It is not compatible with the Pixhawk 5X/6X [42].
RC link	RadioMaster Pocket (ELRS, FCC) + RP1 V2 receiver	$71.50 [47] + $18.99 [48]	ELRS 2.4 GHz, EdgeTX [47]; RP1 2.2 g, CRSF [48]	The Pocket needs two 18650 cells, which are not included [47].
Telemetry	Holybro SiK V3, 915 MHz, 100 mW (pair)	$58.99 [43]	20 dBm; 23.5 g per module [43]	Standard ArduPilot telemetry link.
Remote ID	Dronetag DRI	$55.00 [67]	ArduPilot-compatible; 1.5 g; powered from the flight controller [66]	Dronetag states that the FAA accepted it (DoC RID000000252) [66].
Flight battery	Tattu 5200 mAh 4S 35C LiPo, XT60	$62.99 [49]	436.5 g; 76.96 Wh	In stock at two sellers.

The S500 V2 ARF kit ($259.00, with motors and ESCs [41]) is close in price to the recommended frame plus propulsion ($264.99), but Holybro publishes no thrust table for its motor. The Holybro X500 spare motors, ESCs, props, and PDB were all sold out when checked [44].

Software Architecture

The project uses no custom flight code this semester. It runs stock ArduPilot Copter on the flight controller, with Mission Planner as the ground control station. The work goes into configuration, simulation, tuning, and testing. That keeps M4 within reach and leaves a clean base for next semester's autonomy work.

Layer	This semester	Next semester
Ground	Mission Planner on a laptop, over USB or SiK telemetry [7, 8]	Same
Flight controller	ArduPilot Copter: Auto mission, flight modes, failsafes, geofence, logging	Adds on-board Lua relaunch logic
Companion computer	None; a mount point and mass allowance are reserved	Raspberry Pi Zero 2 W over MAVLink [10]

Autonomous mission (M4)

In Auto mode the copter follows a mission stored in the autopilot [12]. The M4 mission uses these commands:

Order	Command	Behavior
1	NAV_TAKEOFF	Climbs straight up from the current location to the set altitude [11]
2	NAV_WAYPOINT	Flies a straight line to a latitude, longitude, and altitude [11]
3	NAV_RETURN_TO_LAUNCH	The mission equivalent of RTL mode [11]
alt.	NAV_LAND	Lands at the current location or at given coordinates [11]

ArduPilot advises that missions normally end with an RTL command [12]. RTL first climbs to whichever is higher: RTL_ALT_M, or the current altitude plus RTL_CLIMB_MIN_M (named RTL_CLIMB_MIN before Copter 4.7). Near home, RTL_CONE_SLOPE may limit this climb [14, 15]. RTL_ALT_M defaults to 15 m [14], and RTL_CLIMB_MIN_M defaults to 0 m [16]. With the 10 m mission altitude, the vehicle climbs 15 m − 10 m = 5 m before it returns. Over home, RTL pauses for RTL_LOIT_TIME and then lands [14].

When a mission starts on the ground in Auto, by default it begins when the throttle is raised above zero. If the copter is already flying, it begins when Auto is selected [12]. AUTO_OPTIONS can allow arming in Auto and a takeoff without a throttle raise [12]. This semester does not use that option, but next semester's relaunch logic needs it. WP_SPD sets the maximum horizontal mission speed in m/s [12], and its default is 10 m/s [12]. The design calls for a deliberately slow cruise, so the team sets WP_SPD below the default after SITL and M2 tests.

Flight modes and switches

Mode	Role	Key behavior
Loiter	Arming, M2 hold test, pilot recovery	Holds location, heading, and altitude [13]. Arms only with GPS 3D lock and HDOP below 1.4 [13].
Auto	M4 mission	Runs the stored mission [12]
RTL	Mission end, failsafe action	Climbs, returns home, and lands [14]
Land	Failsafe action, pilot landing	Descends straight down [17]. Below 10 m it descends at LAND_SPD_MS [17], which is set in m/s and defaults to 0.5 m/s. Enter 0.5, not 50 [16]. It disarms on touchdown if the throttle is at minimum [17].
AltHold	AutoTune	AutoTune needs a copter that already flies acceptably in AltHold [27]

The transmitter carries switches set through RCx_OPTION: RTL (4), Land (18), Loiter (56), Auto (16), AutoTune (17), Fence enable (11), and Motor Emergency Stop (31) [22]. Motor Emergency Stop stops the motors immediately [22]. The pilot can take control back from Auto at any time with the flight-mode switch [12].

Failsafes and geofence

Trigger	Setting	Action
Low battery: voltage below BATT_LOW_VOLT for more than 10 s [18]	BATT_FS_LOW_ACT = 2; BATT_LOW_VOLT set per pack (see Safety)	RTL, or Land if RTL is not possible [18]
RC link lost for longer than RC_FS_TIMEOUT (default 1 s) [19]	FS_THR_ENABLE = 1	RTL, or Land if the GPS position is not usable [19]
Geofence breach	FENCE_ENABLE = 1 [20]; cylindrical fence centred on home [21]; FENCE_RADIUS = 50 m; FENCE_ALT_MAX above RTL_ALT_M	FENCE_ACTION = 1: RTL or Land [20]

The geofence values are a team decision. ArduPilot says FENCE_RADIUS should normally be at least 50 m [21], so 50 m covers the 30 m waypoint with 20 m to spare. After a breach, a backup fence goes up 20 m farther out, and the vehicle switches to Land 100 m past the fence [20]. The radio failsafe does nothing while disarmed and disarms a landed copter at once [19]. After the RC link returns, the copter stays in its failsafe mode [19]. Fence-breach behavior is checked in SITL before any flight.

Simulation, calibration, and tuning

- **SITL first.** ArduPilot's SITL runs Copter without any hardware [23]. The `sim_vehicle.py` script builds the firmware and starts the simulator [24]. With `--no-mavproxy`, SITL listens on TCP port 5760, and Mission Planner connects over TCP [24]. Every mission, parameter change, and fault path in Figure 2 runs in SITL before it flies.
- **Compass calibration.** Run the onboard calibration in Mission Planner under SETUP, Mandatory Hardware, Compass [25]. Rotate the vehicle so each side points down in turn [25], then reboot before arming [25].
- **CompassMot.** This compensates for interference from power wires, ESCs, and motors [26]. The props are flipped so they push the copter down [26]. A result under 30% is acceptable, 31–60% is a "grey zone", and above 60% means moving the autopilot up and away [26].
- **AutoTune.** AutoTune tunes Stabilize P, Rate P and D, and maximum rotational accelerations [27]. It runs from an auxiliary switch [27], twitches the copter about 20 degrees for a few minutes [27], and saves the gains when the pilot lands and disarms with the switch high [27].

Logging for acceptance tests

Dataflash logs are stored on the autopilot and downloaded after each flight [28]. By default, logging starts at first arm [28], and LOG_BITMASK selects which messages are recorded [28]. The GPS log message includes NSats, HDop, Lat, Lng, and Alt [29]. The BAT message records Volt, Curr, CurrTot, and EnrgTot, and the MODE message records mode changes [29]. The Holybro M10 lists 2.0 m CEP [38], which is the same size as the M4 2 m landing target. So M4 landings are measured with a tape measure, and the GPS log is supporting evidence only.

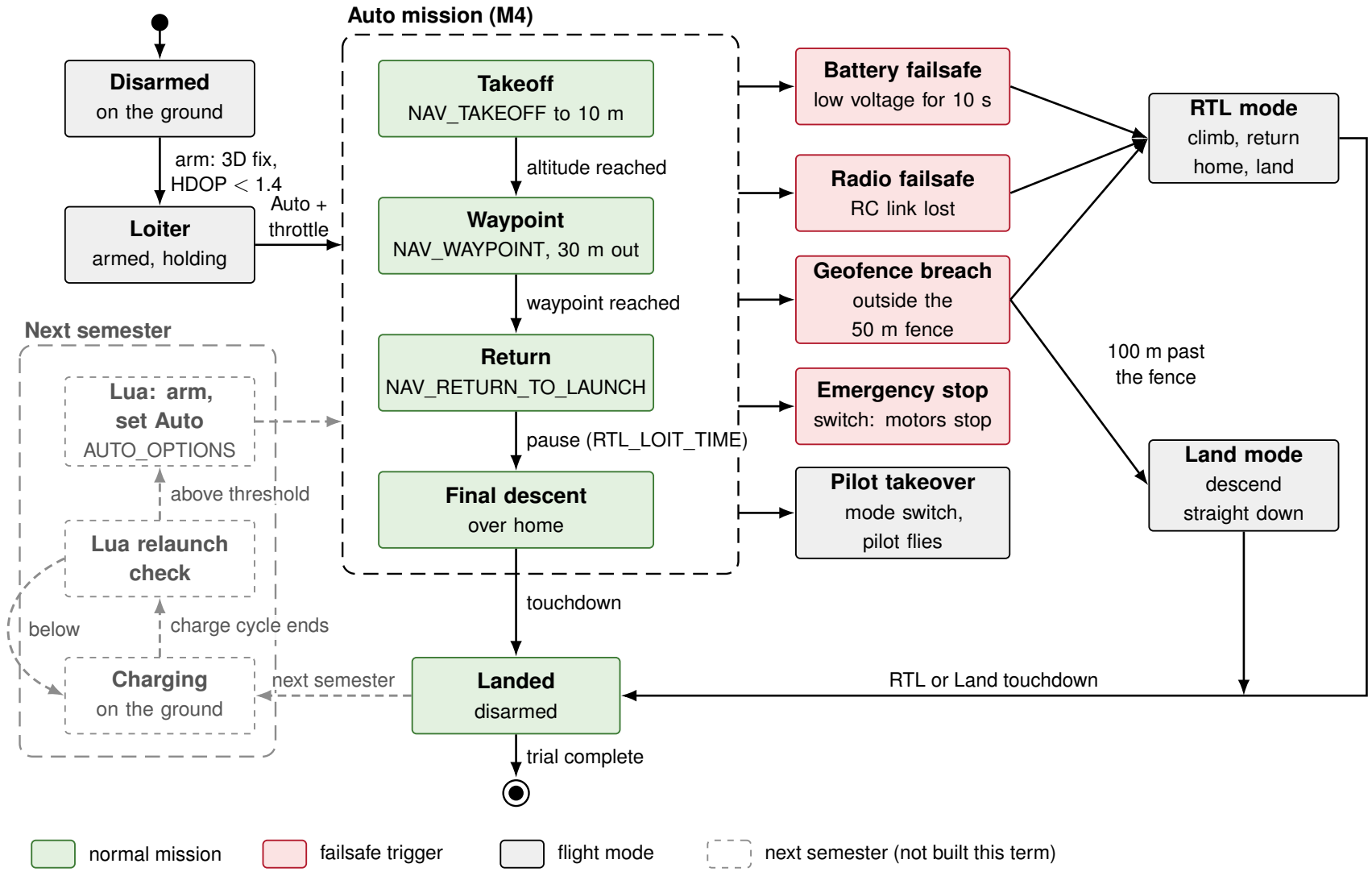


Figure 2. Mission and fault state machine. Everything outside the Next semester block is this semester's scope. The charge and relaunch states are next-semester work.

Next semester: companion computer and relaunch logic

The Raspberry Pi Zero 2 W connects on SERIAL2, where SERIAL2_PROTOCOL = 2 (MAVLink 2) is the default and SERIAL2_BAUD = 921 sets 921600 baud [30]. ArduPilot's Raspberry Pi page says a Pi's 5 V draw can vary from 80 mA to close to 2.5 A, and it does not recommend powering the Pi from the TELEM port [30]. Lua scripts add behavior in a sandbox without changing the core flight code [31]. The bindings `capacity_remaining_pct`, `arm()`, and `set_mode()` exist [31], so the relaunch logic in Figure 2 is feasible. Scripting is turned on with SCR_ENABLE = 1, and scripts go in the SD card's APM/scripts folder [31].

Electrical Design

The project has two power paths. The **flight path** runs from the LiPo through the power module to the ESCs and the flight controller. The **M5 bench-charge path** runs from mains power to the target Li-ion pack.

Flight power distribution

The PM02 V3 accepts 2S–12S and is rated 60 A continuous [42]. It supplies the flight controller at 5.2 V, up to 3 A[†] [42]. The AIR 20A ESCs have no BEC[†] [45], so the 6C and its peripherals run only from the PM02. The Dronetag DRI takes its power from the flight controller at 3 mA average and 10 mA maximum [66].

Gap: power distribution board. Neither the X500 Frame Kit nor the T-Motor set includes a PDB, and the Holybro X500 V2 PDB is sold out [44]. The Holybro PM06 V2 combines a power module with distribution for four ESCs [87] and is the first backup (see Budget). The team chooses an in-stock PDB or harness in M1 and confirms connector fit (XT60 at the PM02 and battery).

Propulsion figures and hover estimate

T-Motor's table for the AIR2216II with the T1045 at 16 V gives [45]:

Throttle	Thrust per motor	Current	Power
45%	373 g	2.86 A	46 W
50%	447 g	3.60 A	58 W
100%	1332 g	16.37 A	260 W

T-Motor states that its bench data are for reference only [45]. The T1045's listed thrust limit is 1.2 kg [45], which is below the 100% row.

In a steady, level hover, the four rotors together carry the vehicle's weight, so each rotor carries about one quarter for a centred CG. With the mass sum in Mechanical Design (1550.8 g with the LiPo; 1674.3 g with the P45B cells) and g = 9.80665 m/s^2 [65], each motor needs 3.80 N (387.7 gf) with the LiPo and 4.10 N (418.6 gf) with the P45B cells. Linear interpolation between the 45% and 50% rows gives:

- **Hover power: about 194 W (LiPo) and about 214 W (P45B pack). Estimate.**
- **Hover current at 16 V: about 12.0 A and 13.3 A. Estimate.**

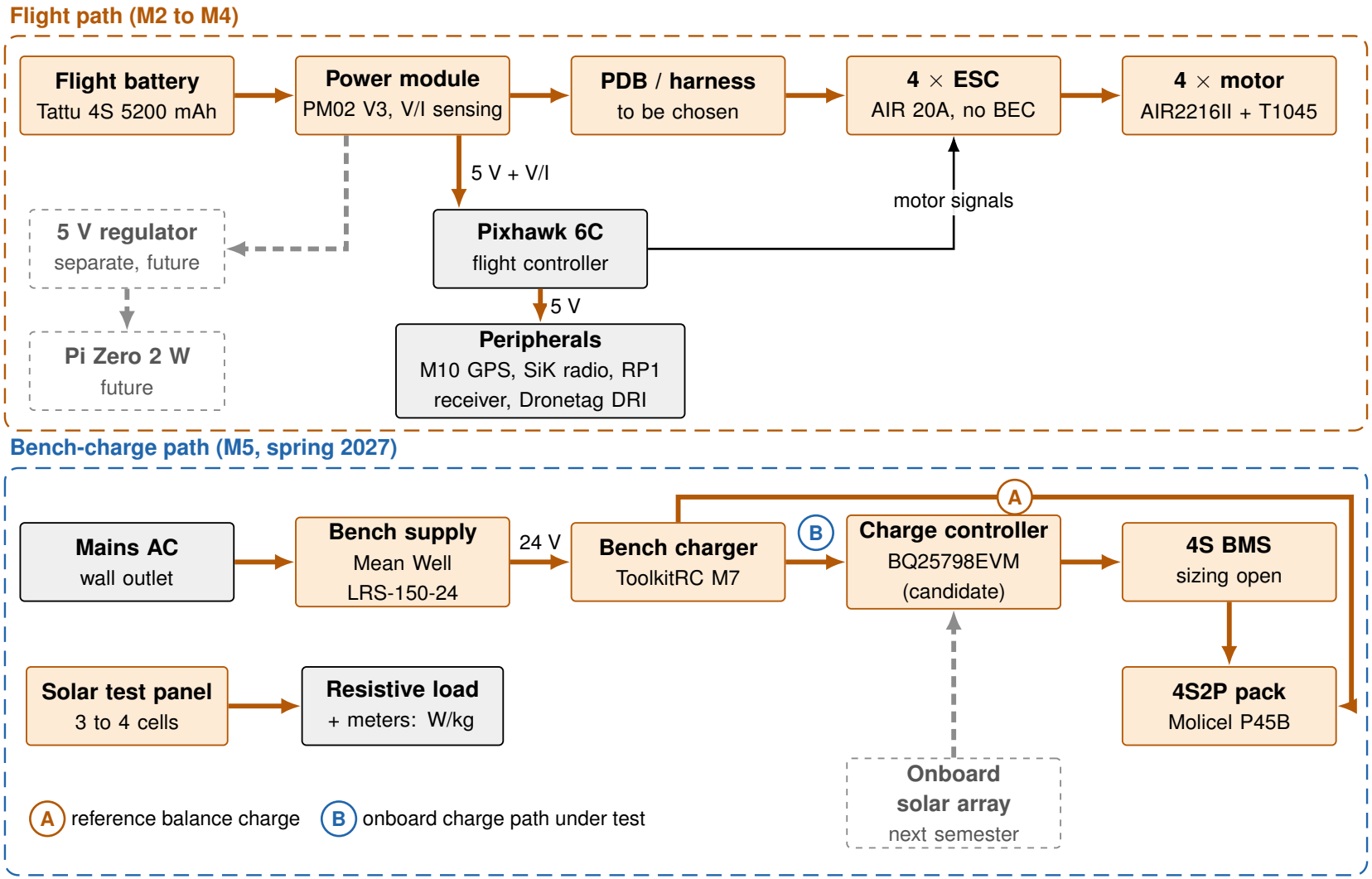


Figure 3. Electrical power paths. Grey dashed arrows and boxes are next-semester items.

These estimates are lower bounds. The mass sum leaves out wiring and fasteners, the data are 16 V bench data, and no efficiency factor is applied. If the existing motors pass inspection and are reused instead, this estimate is redone, because the T-Motor table applies only to the AIR2216II. If the kit ships T1045 II props at the 17 g a retailer lists, the mass rises 16 g and hover power rises about 2 W. M2 measures the real hover power.

Batteries and BMS

Flight battery. The Tattu pack is 5200 mAh, 14.8 V (4S), 35C, with an XT60 plug [49]. It stores 14.8 V $\times$ 5.2 Ah = 76.96 Wh and weighs 436.5 g [49].

Target Li-ion pack (M5). The Molicel P45B is rated 4500 mAh (16.2 Wh) at 3.6 V nominal, with a 4.2 V charge voltage [50]. It allows 45 A maximum continuous discharge with an 80 °C cut-off [50]. Datasheet v1.2 gives 70 g maximum and 242 Wh/kg [50]. A 4S2P pack stores 8 $\times$ 16.2 Wh = 129.6 Wh in at most 8 $\times$ 70 g = 560 g of cells, not counting the BMS, nickel strip, and wiring.

BMS sizing is open. The recommended Battery Hookup 4S BMS lists 30 A continuous discharge, 30 A charge, 50 mA balancing, and a 4.2 V per-cell overcharge cut-off [52]. It is sold out [52]. For bench charging, 30 A is ample. For flight it is too small: the motors can draw 68 A at peak (4 $\times$ 17 A)†. The pack does not fly this year, so BMS sizing is closed before it flies.

Nickel strip. Protomesh has offered 25 cm of nickel strip, enough for eight cells. The links between series groups carry the full pack current: about 12–13 A in hover (estimate, above) and about 65 A at full throttle (4 $\times$ 16.37 A from T-Motor's table [45]). Before use, the team confirms the strip is pure nickel rather than nickel-plated steel, and that its current rating covers the full-throttle load; a single thin strip may need to be widened or doubled.

M5 bench-charge chain

Stage	Part	Key ratings
Mains to DC	Mean Well LRS-150-24	85–132 / 170–264 VAC in, selected by switch; 24 V, 6.5 A, 156 W out [55]
Bench charger and source	ToolkitRC M7	200 W, 10 A; charges LiPo, LiHV, LiFe, Li-ion, and NiMH; 7–28 V DC input [53]
Onboard charge controller (candidate)	TI BQ25798EVM	I^2C-controlled 1–4 cell, 5 A buck-boost solar charger with MPPT; 3.6–24 V input [56]
Interface	TI USB2ANY	Needed to control the EVM from a laptop; not included with the EVM†
Protection	4S BMS	See above

- **Chain A, the reference charge (supply $\rightarrow$ M7 $\rightarrow$ pack).** A conventional balance charge, which meets the Safety rule for series packs.
- **Chain B, the onboard path under test (supply $\rightarrow$ M7 in Power mode $\rightarrow$ BQ25798EVM $\rightarrow$ BMS $\rightarrow$ pack).** The supply's 24 V output sits exactly at the BQ25798's 24 V input limit [56], so the M7 feeds the EVM as a current-limited source set below 24 V†. The 156 W supply caps the M7 below its 200 W rating, so the M7's input power limit is set to 156 W or less.

A 4S pack is full at 4 $\times$ 4.2 V = 16.8 V. The M7's Li-ion profile charges to a fixed 4.10 V per cell (16.4 V for 4S); only its LiPo, LiHV, and LiFe profiles let the user set the cut-off† [54]. The team

chooses the profile at M5. The team charges at 50% or less of the rated current, following UW EH&S [78]: 0.5×4.5 A $= 2.25$ A per cell [50], or 4.5 A for the 2P pack. **Choosing the charge controller.** The BQ25798EVM is a candidate. With the USB2ANY interface its kit costs $335.32; Chain B, the least certain part of this chain, depends on it (the M7 Power-mode use is unverified†); and the EVM was out of stock when checked [57]. Before M3 the team researches a cheaper onboard charger, and the M3 gate chooses between it and the EVM. **What M5 logs:** input and pack voltage and current, pack temperature, per-cell voltages before and after each charge, charge time, energy in, and any fault. The team sets the pass thresholds before M5 testing begins.

Solar test panel: measurement, not prediction

The team builds only a 3–4 cell test panel, in spring 2027. The seller lists each cell at 0.625 V at maximum power [61], so three or four cells in series give about 1.9–2.5 V. That is below the BQ25798's 3.6 V minimum input [56], so the panel does not drive the charger. It is a measurement article:

1. Weigh the finished panel, including laminate and leads.
2. In real sun, sweep a resistive load from open circuit to short circuit. At each step, record panel voltage, current, and cell temperature.
3. Take the maximum-power point, and compute watts per kilogram as measured power divided by measured mass.

The cells are de-branded [61], so the Maxeon Gen III datasheet (3.72 Wp, 24.3%, 0.632 V, 5.89 A for the top bin; about 6.5 g per cell [60]) is a reference only. The Genasun GVB-8 is not recommended: it weighs 185 g and has no standard 4S Li-ion model [59].

Mechanical Design

The airframe is built on a new frame, because the existing frame is misaligned. Other existing parts are reused only if they pass the M1 inspection. The mechanical goals this semester are a stiff, well-balanced quadcopter for M2 and M4, correct CG with both batteries, reserved space and mass for next semester, and a fixture for the solar test panel.

Frame

The Holybro X500 V2 has a 500 mm wheelbase, a carbon-fibre frame, and a listed weight of 610 g [40]. Its body plates are 144×144 mm and 2 mm thick, with 28 mm between them [40]. The arms are 16 mm carbon-fibre tubes, and the landing gear uses 16 mm and 10 mm tubes at 215 mm height [40]. It has two 10 mm $\times$ 250 mm rails, a battery mount with two straps, and a platform board with holes for a GPS and common companion computers [40]. ArduPilot says arm flex is a big cause of vibration and arms should be as rigid as possible [34].

Motor fit. The X500 motor mounts are 16×16 mm and 19×19 mm [40]. We could not verify the AIR2216II hole pattern, because T-Motor's drawing is an image only. The team confirms the fit at M1. ArduPilot says the most common cause of yaw imbalance is a motor that is not vertical on a round arm, and that this is fixed in hardware [35]. The build checks each motor for vertical alignment before the first flight.

Props. Both Holybro frames ship with 1045 props [40, 41], so the 10×4.5 in T1045 fits.

Mass budget

Item	Mass
X500 V2 frame	610 g [40]
4 × AIR2216II motors	256 g (4 × 64 g) [45]
4 × AIR 20A ESCs	84 g (4 × 21 g) [45]
4 × T1045 props	52 g (4 × 13 g) [45]
Pixhawk 6C	34.6 g [37]
M10 GPS	32 g [38]
SiK air unit	23.5 g [43]
PM02 V3	20 g [42]
RP1 V2 receiver	2.2 g [48]
Dronetag DRI	1.5 g [66]
Subtotal, no battery	**1115.8 g**
+ Tattu 4S LiPo (436.5 g [49])	**1552.3 g**
+ 4S2P P45B cells instead (560 g)	**1675.8 g**
Pi Zero 2 W allowance (next semester)	+10.0 g (retailer listing [63])

The sums leave out wiring, connectors, the PDB, mounts, fasteners, the BMS, and the charge electronics. The props may weigh 17 g each (T1045 II) rather than 13 g, which adds 16 g. The M1 weigh-in records the real all-up mass. (The hover estimates above used the subtotal without the 1.5 g Remote ID module; the difference is negligible.)

Battery mounting and centre of gravity

ArduPilot says the autopilot should sit close to the vehicle's centre of gravity, both horizontally and vertically [32]. The battery mounts on a board that clips to the rails [79] and is held by two straps [40].

- **Tattu LiPo.** The pack measures 133 × 45 × 33.5 mm [49], so it can slide fore and aft on the 250 mm rails to trim the CG. Its 5200 mAh is 200 mAh above Holybro's recommended 4S 3000–5000 mAh range [40]. The team confirms the strap fit at M1.
- **4S2P P45B pack.** It is heavier and more compact than the LiPo. A printed PETG tray holds it on the same rail clip, and the team marks a separate balance position for each pack. The P45B pack is bench-charged only, in spring 2027.

Flight controller, GPS, and vibration

The Pixhawk 6C measures 84.8 × 44 × 12.4 mm [37] and has an integrated vibration isolation system [37]. It mounts at the frame centre with its arrow forward [32]. For another orientation, AHRS_ORIENTATION is set before accelerometer calibration [32]. The GPS mast goes on the top plate, at least 10 cm from power wiring and the battery [5]. The X500 build guide fixes the GPS mount with an M3 × 10 screw and an M3 locknut [79].

The team starts without extra dampers and lets the logs decide. ArduPilot treats vibration below 30 m/s/s as normally acceptable; above 60 m/s/s nearly always causes position or altitude hold problems [33]. Accelerometer clipping should ideally be zero, and under 100 is likely acceptable

[33]. If limits are exceeded, the team balances the props [34] and then adds four foam cubes about 3/4 in square [32].

Companion-computer mount and solar test fixture

The Pi Zero 2 W measures 65 × 30 mm [64] and mounts on the platform board. Its plate is designed this semester but left off the aircraft.

The test-panel fixture uses LinkSolar cells, 125 × 125 mm with chamfered corners [61], ready for tabbing wire or conductive adhesive [61]. The cells sit on a PETG backing plate. Prusa rates PETG for most exterior use below 80 °C, and says PLA softens and deforms above 60 °C [80], so PETG is used for sun-exposed parts. TPU is used only for spacer washers, a use Prusa lists for flexible filament [80].

Constraints for next semester's onboard array

1. **No shading, zero added mass for looks, no loss of serviceability** (team design rule).
2. **Prop discs.** If the array sits below the prop plane, area far from the centre lies under a prop disc. The team studies cut cells, a raised mount, or an accepted overlap in M6.
3. **Mass.** Array mass adds to the 1675.8 g sum, and the thrust budget is rechecked with the measured watts per kilogram.
4. **Magnetic interference.** Array wiring follows the 10 cm GPS rule [5], and CompassMot is re-run after installation [26].

Project Goals

What success looks like this semester

ID	When	Goal	Acceptance criterion	Evidence
M1	Sep 28–Oct 11	Platform and compliance	Existing parts inspected and the rest ordered; EH&S decision on in-house pack building recorded; geofence set to FENCE_RADIUS = 50 m and FENCE_ALT_MAX above RTL_ALT_M; compliance items in the Safety section complete	Inspection record; parts list with costs; saved parameter file; compliance checklist
M2	Oct 12–Nov 8	Reliable position hold; hover power measured	Hands-off Loiter for 3 min within a 2 m radius, on three flights. Hover power measured.	Logs: MODE (no mode change in the hold), GPS Lat/Lng (radius), BAT Volt × Curr (hover power)
M3	Nov 30–Dec 4	Decision gate	Memo: choose the M5 onboard charge controller and pack approach, and set the onboard-array direction for spring	Decision memo

ID	When	Goal	Acceptance criterion	Evidence
M4	Nov 9–20	Autonomous mission (**primary deliverable**)	Takeoff, 30 m waypoint at 10 m altitude, return, landing within 2 m, five consecutive trials	Five logs (CMD, MODE, GPS); tape measurement of each landing
M5	Spring 2027	Onboard charging electronics bench-tested	The target Li-ion pack charges through the onboard charge path from a bench source, logged. Pass thresholds set by the team before testing.	Bench log
M6	Spring 2027	Next-semester readiness	M5 results and the test-panel watts-per-kilogram result combined into a design for the onboard array and the full land–charge–relaunch loop	Design document
M7	Spring 2027 (fall handoff Dec 7–12)	Handoff package (**never cut**)	Parameters, logs, documentation, failure record delivered	The package

How each criterion is measured

- **Hands-off Loiter (M2).** A MODE log with no mode change during the 3 minutes shows the hold was hands-off [29]. The team computes the largest horizontal distance of any GPS fix from the mean position. The hold passes if that distance is 2 m or less. Loiter quality depends on GPS lock, compass interference, and vibration [13], so a failed hold points to those three areas first.
- **Hover power (M2).** Power = BAT Volt $\times$ BAT Curr, averaged over the steady part of the hold.
- **Autonomous mission (M4).** The CMD log shows the mission that was loaded, and the GPS track shows what flew [29]. The MODE log shows Auto with no pilot takeover.
- **Landing within 2 m (M4).** Measured with a tape measure from the marked takeoff point. The GPS log is supporting evidence only, because the M10's listed 2.0 m CEP is the same size as the target.
- **Five consecutive trials (M4).** Any aborted trial resets the count and goes in the failure record.

Descope rules and the pause point

- If M2 slips past November 8, M4 moves to November 30–December 4, and M3 moves to finals week.
- If M4 slips past December 4, it becomes the first milestone of spring, ahead of M5.
- If M5 slips in spring, M6 is dropped.
- The handoff package is never cut: an interim one at the end of fall, and the full M7 in spring.

At the minimum, the fall ends with a vehicle that flies the M4 mission and a handoff package, so spring starts from a flying, documented aircraft.

Stages and future expansion

- **Stage 1 (fall 2026).** Autonomous flight.
- **Stage 2 (spring 2027).** The onboard charge path bench-tested, the solar test panel measured, and the array and full-loop design.
- **Stage 3 (after spring).** The onboard solar array, the full land–charge–relaunch loop, the Raspberry Pi Zero 2 W companion computer, and Lua relaunch logic.
- **Out of scope this semester.** Unattended charging, beyond-visual-line-of-sight or unsupervised flight, vision-based landing-site selection, and mission-level autonomy.

Applications. Persistent monitoring, disaster damage survey, and an aerial Meshtastic mesh relay. Commercial Meshtastic devices already strap onto drones to make aerial relay stations [81]. A survey preprint lists limited flight endurance among the challenges for UAV-based relays [82]. Self-recharging is the part this project addresses.

Prospective partner. The Protomesh Collective's GitHub page describes it as an antiprofit formation that builds low-cost Meshtastic radios, hosts regular introductory Meshtastic and LoRa workshops, and aims to make its designs freely available whenever possible [83]. Its support is offered on the condition that RAS investigates low-cost, high-quality quadcopters and open-sources its findings. After a September 25 call, it has offered a Raspberry Pi Zero 2 W and nickel strip, and has donated a spot welder for battery-pack building to Prototype x Oakland. Protomesh has also asked that the project stay civilian; the team will write an ethics statement with them.

Safety

The two main risks are the aircraft and its lithium batteries. The Safety & Compliance Officer owns this section and turns it into written procedures in M1.

Federal rules: the legal path

The recreational exception (49 U.S.C. 44809) applies only if every condition in 44809(a) is met: strictly recreational purpose; a community-based organization's safety guidelines; visual line of sight; giving way to manned aircraft; prior authorization in Class B, C, D, or airport Class E airspace; at most 400 ft above ground in Class G; a passed knowledge and safety test; and registration and marking [70]. Other operations must follow the general UAS rules [70]. A statutory note says a "recreational purpose" includes a UAS operated by an institution of higher education for educational or research purposes [70].

Part 107 (14 CFR Part 107) covers small unmanned aircraft under 55 lb [71]. A person may fly if they hold a remote pilot certificate and meet the §107.65 recency rule, or if they are under the direct supervision of a remote pilot in command (RPIC) who can take direct control at once [71]. The RPIC must hold the certificate [71]. An applicant must be at least 16, and one way to qualify is an initial knowledge test [71]. Main limits [71]:

- The RPIC, the person at the controls, and any visual observer must be able to see the aircraft throughout the flight, with vision unaided except by corrective lenses.
- ATC authorization in Class B, C, D, or airport Class E airspace.

- 400 ft above ground level (about 122 m), except near structures; 87 knots (100 mph) groundspeed.
- No flight over people unless an exception applies; yield right of way to all aircraft.
- Before flight: assess the environment, brief participants, and confirm the control links work.

Recommendation: fly under Part 107, with a certificated remote pilot as Pilot in Command on every flight. The statute does not say whether a student club project counts as "operated by an institution of higher education". Pitt's drone policy encourages all flyers to follow Part 107 and calls the exception a viable option for educational or research flights [74]. Part 107 does not depend on how the club is classified. The Safety & Compliance Officer confirms this path with the University in M1.

Registration. Registration and marking are the usual way to meet 14 CFR 48.15 [72]. The only weight exception is for aircraft flown exclusively under 44809 and weighing 0.55 lb (about 249.5 g) or less [72]. Under Part 107, registration is required at any weight. The mark goes on an external surface [72]. The fee is $5.00 per aircraft [72].

Remote ID. Except as authorized by the FAA or under §89.120, an unmanned aircraft may fly only as a standard Remote ID aircraft or under the §89.115 alternatives [73]. The home-built exemption covers only design and production rules [73]. With a broadcast module, the module's serial number must be on an FAA-accepted declaration of compliance, the pilot checks it works before takeoff, keeps the aircraft in sight, and lands as soon as practicable if the broadcast stops [73]. The other option is to fly with both aircraft and pilot inside an FAA-recognized identification area [73]. **Recommendation:** fit the Dronetag DRI, which Dronetag states the FAA accepted (DoC RID000000252) [66]. The Safety & Compliance Officer confirms the DoC on the FAA site in M1.

University of Pittsburgh rules

Policy RI 13 (Drones), effective March 27, 2023, covers drone use on University-owned or leased property [74]. It defines University Directed Flights as educational, research, or commercial flights by faculty, staff, students, or contractors [74]. We read a student project as a Directed Flight and confirm this with the University in M1. The rules are:

- Notify the University at least 24 hours before each outdoor flight on University property [74], on the Outdoor Directed Drone Flight Notification Form. ORSTC then informs University Police [75].
- No flight inside University buildings without an exception [74]; indoor requests go in at least five business days ahead [75].
- No recording where people have a reasonable expectation of privacy [74]. University Police may stop any unsafe or noncompliant flight [74].
- Contact: drones@pitt.edu [75].

Pitt EH&S Guideline 02-009 [76] covers Li-ion and LiPo cells and packs:

- §2.1: remote-controlled vehicles and devices may not be operated, stored, or recharged inside University buildings. The guideline's introduction exempts approved University operations and faculty-directed academic or research activities. **Where RAS may charge and store packs is**

confirmed with EH&S in M1.

- §2.3: packs must be UL-certified and from a reputable manufacturer. §2.5: never solder cells or packs.
- §5.2–5.3: research devices need a hazard analysis and written SOPs.

Flight operations

Five team members cover six lead roles (Project Lead, Flight & Autonomy, Power & Charging, Mechanical & Fabrication, Safety & Compliance, Documentation & Design); each lead has a backup. On flight days the roles are Pilot in Command, Safety Spotter, and Logger. The person flying is never the Safety Officer.

Pre-flight checklist (team procedure):

1. Assess the site, weather, airspace, and people nearby; file the University notice at least 24 h ahead.
2. Brief the crew on roles, emergency procedures, and hazards.
3. Inspect the battery for damage or swelling and check cell voltages.
4. Check the control link and the Remote ID module; check geofence and failsafe parameters against the approved file.
5. Have a fire extinguisher at the flight line.

The failsafe, geofence, and emergency-stop settings are in Software Architecture. The fence ceiling sits just above RTL_ALT_M (default 15 m), far below the 400 ft limit. After a crash, the pilot cuts motor power, and the team watches the pack before handling it, because a delayed fire can occur hours or days after an impact [76].

Battery safety

Topic	Practice	Source
Charge voltage	Not above 4.2 V per cell for most batteries	[76]
Charger	Balance-charge series packs with a charger that monitors each cell; never parallel-charge	[76]
Supervision	Constantly attended while charging	[76, 77]
Surface and container	Heat-resistant, noncombustible, nonconductive surface; charge and store in a fire-retardant container	[76]
Before and after charging	Do not charge a pack still warm from use; disconnect when charging ends; plug chargers straight into wall outlets	[76]
Storage	60–70% of capacity, about 3.8 V per cell, in fire-resistant containers; Class ABC or CO2 extinguisher at the storage area	[76]
Low-voltage limit	Do not over-discharge below 3 V	[76]
Damaged pack	Disconnect if it overheats, smells, or changes shape; place a damaged pack in a metal can with sand	[76]
Fire	Evacuate and call 412-624-2121; EH&S is 412-624-9505	[76]

BATT_LOW_VOLT. The ArduPilot default is 10.5 V [18], which is 2.625 V per cell on a 4S pack, below Pitt's 3 V floor. The team sets BATT_LOW_VOLT to at least 12.0 V (4×3.0 V) and chooses the final value from voltage logged under load in M2, so normal sag does not trip the failsafe early.

M5 bench charging. A session runs only while the Power & Charging Lead or backup is present. Charging stays inside the P45B's 0–60 °C charge range [50], at 50% or less of the rated current (2.25 A per cell) [78, 50]. The P45B's discharge cut-off is 2.5 V and its maximum charge current is 13.5 A with a 70 °C cut-off [50]. Work is done on non-conductive surfaces with a CO2 or ABC extinguisher at hand [77]. Failure scenarios, including thermal runaway, are considered during design and testing [78].

Pack building. CMU EHS says to consult EHS when building battery systems, wear PPE, and work behind a blast shield [77]. UW EH&S says battery system design needs qualified designers [78]. Pitt forbids soldering cells [76]. **Recommendation:** buy a pre-assembled UL-certified pack unless Pitt EH&S approves an in-house build in writing. This is an M1 gate: the loose cells and BMS in the Budget are bought, and the donated nickel strip used, only after written EH&S approval. Without approval, the team buys a pre-assembled UL-certified pack instead, priced at M1.

Hazards, controls, and owners

Hazard	Controls	Owner
Wrong legal path	Part 107 with a certificated PIC; confirmed with the University in M1	Safety & Compliance Officer
Unregistered aircraft or no Remote ID	Register and mark; Dronetag DRI checked before each flight	Safety & Compliance Officer
Flight without University notice	Notice filed at least 24 h ahead	Safety & Compliance Officer
Fly-away or loss of control	Geofence; RC and battery failsafes; mode-switch takeover; SITL first	Flight & Autonomy Lead; PIC
Prop strike on people	No overflight of non-participants; Safety Spotter; emergency stop	PIC; Safety Spotter
Manned aircraft conflict	Visual line of sight; yield right of way; under 400 ft	PIC; Safety Spotter
Battery fire while charging	Attended balance charging in a fire-retardant container on a noncombustible surface; extinguisher	Power & Charging Lead
Charging or storage in buildings	EH&S 02-009 §2.1; exception confirmed in M1	Safety & Compliance Officer
Over-discharge in flight	BATT_LOW_VOLT at 3.0 V per cell or higher; battery failsafe	Flight & Autonomy Lead
Damaged pack after a crash	Cut power; watch for delayed fire; inspect; isolate in a metal can with sand	PIC; Logger
Short circuit in pack building	No soldering of cells; EHS consulted; PPE and blast shield	Mechanical & Fabrication Lead
Privacy complaint	No recording where people expect privacy	Project Lead

Timeline

Pitt's fall 2026 term runs August 24 to December 12, with Thanksgiving recess from November 22 to 29; the team confirms these dates against the registrar's calendar. The project starts Monday, September 28, which leaves eight working weeks before Thanksgiving and one after it before finals. Autonomous flight (M1–M4) is scheduled for fall, with M4 due before Thanksgiving, because outdoor flight testing gets harder in December. The charging work and the final handoff move to spring 2027. Work runs simulation-first: SITL runs Copter without hardware [23], so mission logic is rehearsed while parts are in transit. The team is five people, so the weekly load is kept light.

Week (dates)	Milestone	Main tasks	Lead role	Exit check
1 (Sep 28–Oct 4)	M1	Inspect the existing quadcopter's parts; finalize and order the rest. Install SITL; connect Mission Planner. Start the compliance checklist (Part 107 certificate, registration, University notice path, EH&S charging location, EH&S decision on pack building).	Project Lead; Flight & Autonomy; Safety & Compliance	Orders placed; SITL runs
2 (Oct 5–11)	M1	Receive parts. Plan the M4 mission and load failsafe, geofence, and e-stop parameters in SITL. Fall break Oct 9–11.	Project Lead; Flight & Autonomy	Simulated mission completes
3 (Oct 12–18)	M2	Assemble and wire the airframe; mount the GPS on its mast; compass calibration; bench-check failsafes and e-stop.	Mechanical & Fabrication; Flight & Autonomy	Bench arm test passes
4 (Oct 19–25)	M2	CompassMot; logging on; first low hovers in AltHold.	Flight & Autonomy	CompassMot recorded; AltHold flyable
5 (Oct 26–Nov 1)	M2	AutoTune; first Loiter flights.	Flight & Autonomy	Gains saved; Loiter holds
6 (Nov 2–8)	M2	Loiter acceptance flights; measure hover power. **Descope check 1.**	Flight & Autonomy; Power & Charging	**M2 met**
7 (Nov 9–15)	M4	Re-run M4 in SITL; first autonomous trials, geofence on.	Flight & Autonomy	One clean mission
8 (Nov 16–20)	M4	Five consecutive trials.	Flight & Autonomy; Safety & Compliance	**M4 met (primary deliverable)**
9 (Nov 23–29)	—	Thanksgiving recess; no work planned.	—	—
10 (Nov 30–Dec 4)	**M3**	Gate memo: charge controller, pack approach, and array direction for spring. **Descope check 2.**	Project Lead; Power & Charging	Memo reviewed by club leadership
11 (Dec 7–12)	Fall handoff	Archive parameters, logs, and the failure record; write the spring plan.	Project Lead; Documentation & Design	Fall handoff delivered

Week (dates)	Milestone	Main tasks	Lead role	Exit check
Spring 2027	M5, M6, M7	Charging bench tests and the test panel (M5); array and full-loop design (M6); final handoff (M7).	Power & Charging; Project Lead; Documentation & Design	Dates set from the spring calendar

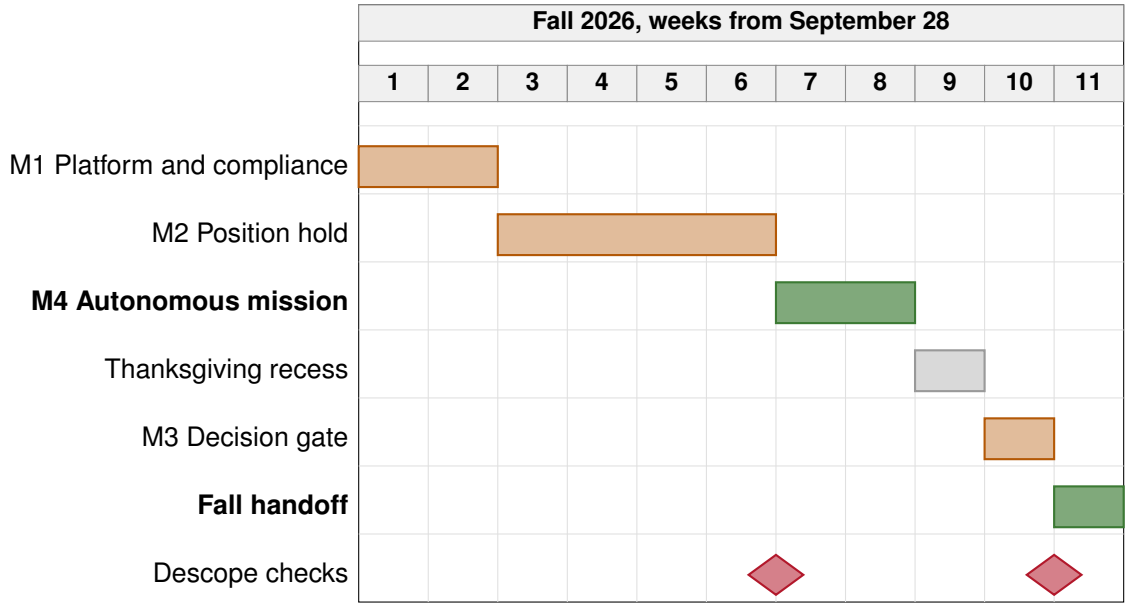


Figure 4. Fall 2026 milestones by week, counting from September 28. M4, the primary deliverable, is due before Thanksgiving recess, and the fall handoff closes the term; diamonds mark the two descope checks. M5 to M7 follow in spring 2027.

Budget

RAS has committed $1,000 for new parts this semester, and more next semester (amount not yet set). RAS has said the list may go over $1,000 and that it can move items between semesters. This proposal shows the full list and makes no cuts. All figures exclude shipping and tax. Recheck every price before purchase.

Item	Qty	Unit	Line total	Status
M4 autonomous flight (fall 2026)				
Holybro Pixhawk 6C (plastic) [37]	1	$199.00	$199.00	
Holybro M10 GPS [38]	1	$43.99	$43.99	
Holybro X500 V2 Frame Kit [40]	1	$135.00	$135.00	
T-Motor Air Gear 450 II combo [45]	1	$129.99	$129.99	
Holybro PM02 V3 [42]	1	$18.99	$18.99	
Holybro X500 V2 PDB (placeholder) [44]	1	$8.59	$8.59	Sold out; replace with an in-stock PDB
Holybro SiK V3 915 MHz, pair [43]	1	$58.99	$58.99	

Item	Qty	Unit	Line total	Status
RadioMaster Pocket, ELRS, FCC [47]	1	$71.50	$71.50	
Molicel P28A 18650 (transmitter) [51]	2	$9.95	$19.90	Sale price; out of stock
RadioMaster RP1 V2 [48]	1	$18.99	$18.99	
Tattu 5200 mAh 4S LiPo [49]	1	$62.99	$62.99	In stock
Dronetag DRI Remote ID [67]	1	$55.00	$55.00	
GT Power LiPo Guard Bag [69]	1	$10.69	$10.69	Recheck listing
FAA registration fee [72]	1	$5.00	$5.00	
M4 subtotal			**$838.62**	
M5 charging build (spring 2027)				
Molicel P45B 21700 [51]	8	$15.95	$127.60	After EH&S approval; sale price (regular $18.95)
Battery Hookup 4S 30 A BMS [52]	1	$16.00	$16.00	After EH&S approval; sold out; sizing open
Pure nickel strip, 25 cm (from Protomesh)	1	$0.00	$0.00	Donated; current rating to confirm; after EH&S approval
Cell spacers, 3D-printed PETG	8	$0.00	$0.00	Printed in-house
ToolkitRC M7 charger [53]	1	$39.99	$39.99	
Mean Well LRS-150-24 [55]	1	$20.39	$20.39	Low stock
LinkSolar 125 mm cells, 10 pcs [61]	1	$46.60	$46.60	
M5 charging build subtotal			**$250.58**	
M5 charge controller (chosen at M3)				
TI BQ25798EVM [57]	1	$199.20	$199.20	0 in stock; 1 due 2026-09-29
TI USB2ANY interface† [58]	1	$136.12	$136.12	Needed for the EVM†
Charge-controller subtotal			**$335.32**	
Later (onboard array stage)				
Raspberry Pi Zero 2 W [62]	1	$0.00	$0.00	Donated by Protomesh

Total	Amount
M4 autonomous flight	**$838.62**
Left of the $1,000 fall commitment after M4	$161.38
M5 charging build (spring)	$250.58
M5 charge controller (spring, chosen at M3)	$335.32
Both semesters, full list	**$1,424.52**
Both semesters without the charge-controller kit	$1,089.20
Later: onboard array (not yet priced)	Pi Zero 2 W donated

Autonomous flight (M4) costs $838.62 in fall and fits inside the $1,000 commitment, with $161.38 to spare. The M5 charging items move to spring, when RAS has said more funding is available. The largest of them is the charge-controller kit: the BQ25798EVM and its USB2ANY interface cost $335.32, about 24% of the full list. That kit is also the least certain part of the design (see Electrical Design), so the M3 gate decides whether to buy it or one of the cheaper options below.

Without the kit, the full list is $1,089.20. The nickel strip and Pi Zero 2 W are donated by Protomesh and the cell spacers are 3D-printed, so none of them are purchases. Any existing part that passes the M1 inspection also comes off the list, and the pack parts are bought only after EH&S approval (see Safety).

The two largest single items are the BQ25798EVM ($199.20) and the Pixhawk 6C ($199.00). Buying eight P45B cells at $15.95 is $127.60; ten at the $15.50 quantity price is $155.00 [51].

Backup parts. If a chosen part is out of stock or delayed, these are the first substitutes. Backups sourced only from marketplace listings are checked with the seller before ordering.

Chosen part	Backup	Notes
Pixhawk 6C	Holybro Pixhawk 6C Mini	Same Holybro family; ArduPilot lists both [84]. Confirm at least 2 MB of flash before buying.
PM02 V3 + X500 V2 PDB	Holybro PM06 V2, $26 [87]	Power module with distribution for four ESCs, 5.2 V / 3 A output, 70 A continuous board rating, compatible with the 6C. Its pre-soldered XT60 and 12 AWG leads are rated 30 A continuous and 60 A burst, so they are upgraded for full-throttle bursts [87].
BQ25798EVM	BQ25792EVM, TI's sibling board [85]	Fine for bench charging, but the BQ25798's main improvement over it is true solar MPPT [86]. Mouser reports a long lead time on this EVM family [85].
BQ25798EVM	Custom BQ25798 board	Protomesh's PCB fabrication makes a small board around the chip realistic. TI also lists a USB-PD evaluation module that carries the BQ25798 [56].
BQ25798EVM	Unbranded 5 A 3S/4S MPPT solar board, $31.50 [88]	Low cost, but unverified; bench-tested before it is trusted.
Battery Hookup 4S BMS	DALY 4S Li-ion 60 A BMS with balance leads and temperature sensor [89]	Choose the Li-ion (3.7 V) version; DALY also sells a 4S LiFePO4 (3.2 V) board [90] that would under-charge these cells. 60 A sits just below the 65 A full-throttle burst, so its rating is confirmed before the pack flies.
X500 V2 frame	Holybro S500 V2 frame kit, $42.00 [41]	No published frame-only mass.
T-Motor Air Gear 450 II	Holybro S500 V2 ARF kit, $259.00 [41]	Includes motors and ESCs, but no published thrust table.
Holybro M10 GPS	Holybro M9N, $54.99 [39]	Lists 1.5 m CEP.
Dronetag DRI	Holybro Remote ID module [68]	Its FAA declaration of compliance is confirmed before flight.
Molicel P28A (transmitter)	18650 cells from 18650BatteryStore.com	The official store Protomesh uses. Imitation sites copy its listings, and their cells may be counterfeit or unsafe [91].
Tattu 4S LiPo	The club's existing LiPo packs	Only packs that pass the swelling and cell-balance check.

Not yet priced: a pre-assembled UL-certified pack (if EH&S does not approve an in-house build), a fire extinguisher, an in-stock PDB or harness, XT60/XT30 connectors and wire, a mains cord for the

bench supply, spare propellers, and the onboard solar array.

References

A dagger (†) marks a figure not yet independently verified; the team rechecks each one before relying on it. All prices were listed on 2026-09-23 or 2026-09-24; recheck every price before purchase. Verification status and open questions for the team are in `ledger/verification-report.md`.

[1] ArduPilot — Pixhawk wiring quick start: https://ardupilot.org/copter/docs/common-pixhawk-wiring-and-quick-start.html

[2] ArduPilot — 3DR power module: https://ardupilot.org/copter/docs/common-3dr-power-module.html

[3] ArduPilot — Power modules: https://ardupilot.org/copter/docs/common-powermodule-landingpage.html

[4] ArduPilot — Connect ESCs and motors: https://ardupilot.org/copter/docs/connect-escs-and-motors.html

[5] ArduPilot — Installing a u-blox GPS/compass: https://ardupilot.org/copter/docs/common-installing-3dr-ublox-gps-compass-module.html

[6] ArduPilot — RC systems: https://ardupilot.org/copter/docs/common-rc-systems.html

[7] ArduPilot — Mission Planner overview: https://ardupilot.org/planner/docs/mission-planner-overview.html

[8] ArduPilot — Connect Mission Planner to the autopilot: https://ardupilot.org/planner/docs/common-connect-mission-planner-autopilot.html

[9] ArduPilot — SiK telemetry radio: https://ardupilot.org/copter/docs/common-sik-telemetry-radio.html

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[11] ArduPilot — Mission command list: https://ardupilot.org/copter/docs/mission-command-list.html

[12] ArduPilot — Auto mode: https://ardupilot.org/copter/docs/auto-mode.html

[13] ArduPilot — Loiter mode: https://ardupilot.org/copter/docs/loiter-mode.html

[14] ArduPilot — RTL mode: https://ardupilot.org/copter/docs/rtl-mode.html

[15] ArduPilot source — mode_rtl.cpp: https://raw.githubusercontent.com/ArduPilot/ardupilot/master/ArduCopter/mode_rtl.cpp

[16] ArduPilot source — config.h: https://raw.githubusercontent.com/ArduPilot/ardupilot/master/ArduCopter/config.h

[17] ArduPilot — Land mode: https://ardupilot.org/copter/docs/land-mode.html

[18] ArduPilot — Battery failsafe: https://ardupilot.org/copter/docs/failsafe-battery.html

[19] ArduPilot — Radio failsafe: https://ardupilot.org/copter/docs/radio-failsafe.html

[20] ArduPilot — Geofencing: https://ardupilot.org/copter/docs/common-geofencing-landing-page.html

[21] ArduPilot — Cylindrical fence: https://ardupilot.org/copter/docs/common-ac2_simple_geofence.html

[22] ArduPilot — Auxiliary functions: https://ardupilot.org/copter/docs/common-auxiliary-functions.html

[23] ArduPilot — SITL simulator: https://ardupilot.org/dev/docs/sitl-simulator-software-in-the-loop.html

[24] ArduPilot — Using SITL: https://ardupilot.org/dev/docs/using-sitl-for-ardupilot-testing.html

[25] ArduPilot — Compass calibration: https://ardupilot.org/copter/docs/common-compass-calibration-in-mission-planner.html

[26] ArduPilot — Advanced compass setup (CompassMot): https://ardupilot.org/copter/docs/common-compass-setup-advanced.html

[27] ArduPilot — AutoTune: https://ardupilot.org/copter/docs/autotune.html

[28] ArduPilot — Downloading and analyzing logs: https://ardupilot.org/copter/docs/common-downloading-and-analyzing-data-logs-in-mission-planner.html

[29] ArduPilot — Log messages: https://ardupilot.org/copter/docs/logmessages.html

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[32] ArduPilot — Mounting the flight controller: https://ardupilot.org/copter/docs/common-mounting-the-flight-controller.html

[33] ArduPilot — Measuring vibration: https://ardupilot.org/copter/docs/common-measuring-vibration.html

[34] ArduPilot — Vibration damping: https://ardupilot.org/copter/docs/common-vibration-damping.html

[35] ArduPilot — Thrust loss and yaw imbalance: https://ardupilot.org/copter/docs/thrust_loss_yaw_imbalance.html

[36] u-blox — NEO-M8 datasheet (R14): https://content.u-blox.com/sites/default/files/NEO-M8-FW3_DataSheet_UBX-15031086.pdf ; Holybro M8N: https://holybro.com/products/m8n-gps

[37] Holybro — Pixhawk 6C: https://holybro.com/products/pixhawk-6c

[38] Holybro — M10 GPS: https://holybro.com/products/m10-gps

[39] Holybro — M9N GPS: https://holybro.com/products/m9n-gps

[40] Holybro — X500 V2 kits: https://holybro.com/products/x500-v2-kits

[41] Holybro — S500 V2 kit: https://holybro.com/products/s500-v2-kit

[42] Holybro — PM02 V3: https://holybro.com/products/pm02-v3-12s-power-module

[43] Holybro — SiK telemetry radio V3: https://holybro.com/products/sik-telemetry-radio-v3

[44] Holybro — X500 V2 spare parts: https://holybro.com/products/spare-parts-x500-v2-kit

[45] T-Motor — Air Gear 450 II combo: https://store.tmotor.com/product/air-gear-450II-combo-set.html

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